Introduction

The dynamics of waves on the surface of water that have very small wavelength are dominated by the surface tension of water. Since surface tension acts as the restoring force, by determining the wavelength for different frequencies, we can determine the surface tension.

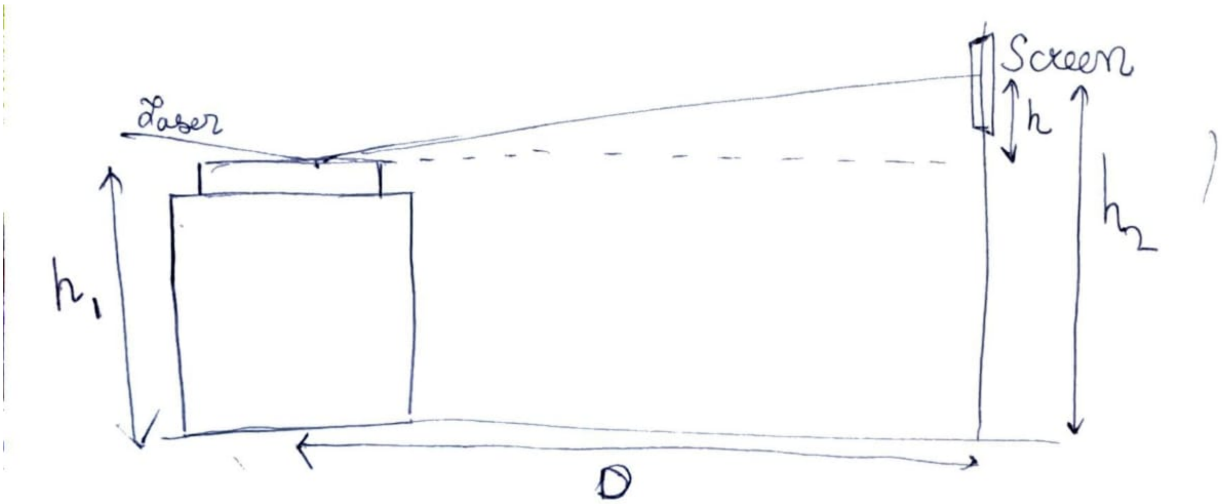
In this experiment, we do reflecting a laser from the surface of water that has the waves and studying the diffraction pattern formed by it.

Experiment Setup and Procedure

Apparatus

A speaker and a device to produce vibrations of known frequency, a container with water, small strip of plastic transferring the vibrations from the speaker to the surface of water, a laser source and stand for it, screen.

Experiment Setup



D is the distance between the screen and the laser.
α is the angle of inclination between the laser and the surface of water.
$h = h_2 - h_1$ is the height difference between the central maxima of the diffraction pattern and the surface of water.

Procedure

The frequency of the speaker is fixed to some value, then the diffraction pattern is allowed to stabilize. Since the frequencies are low and the vibrations are low amplitude, any external disturbances caused by sound smudge the diffraction pattern. So, while avoiding making any sound, we let the diffraction pattern on the screen stabilize and mark the position of the central bright spot and the two spots around. The distance between the spots is measured. Using this information, the surface tension is evaluated.

Formulae and Relations

θ = the diffraction angle.
d = distance between the two first order maximas
$\therefore \theta = d/2D$

The wave number of the capillary waves = $k = 2\pi\ sin\theta\ sin\alpha/\Lambda$ ... (i)
where $\Lambda =$ wavelength of the laser light

The relation between the angular frequency (ω) and wave number (k) is given by

$$\omega^2 = (\frac{\sigma}{\rho})k^3 \ldots (ii)$$

where σ = surface tension of water and ρ is the density of water.

We have $\omega = 2\pi f$
where f is the frequency of the waves, that is same as the frequency of sound generated by the speaker.

$sin\alpha = h/D$
$sin\theta = d/2D$
these values along with (i) in (ii), we get

$$f^2 = (\frac{2\pi\sigma h^3}{\rho\Lambda^3 D^6})\ d^3$$

Plotting f^2 vs d^3, we get the slope of best fit line and finally,

$$\sigma = \text{slope} \times \frac{\rho\Lambda^3 D^6}{2\pi h^3}$$

Data and Analysis

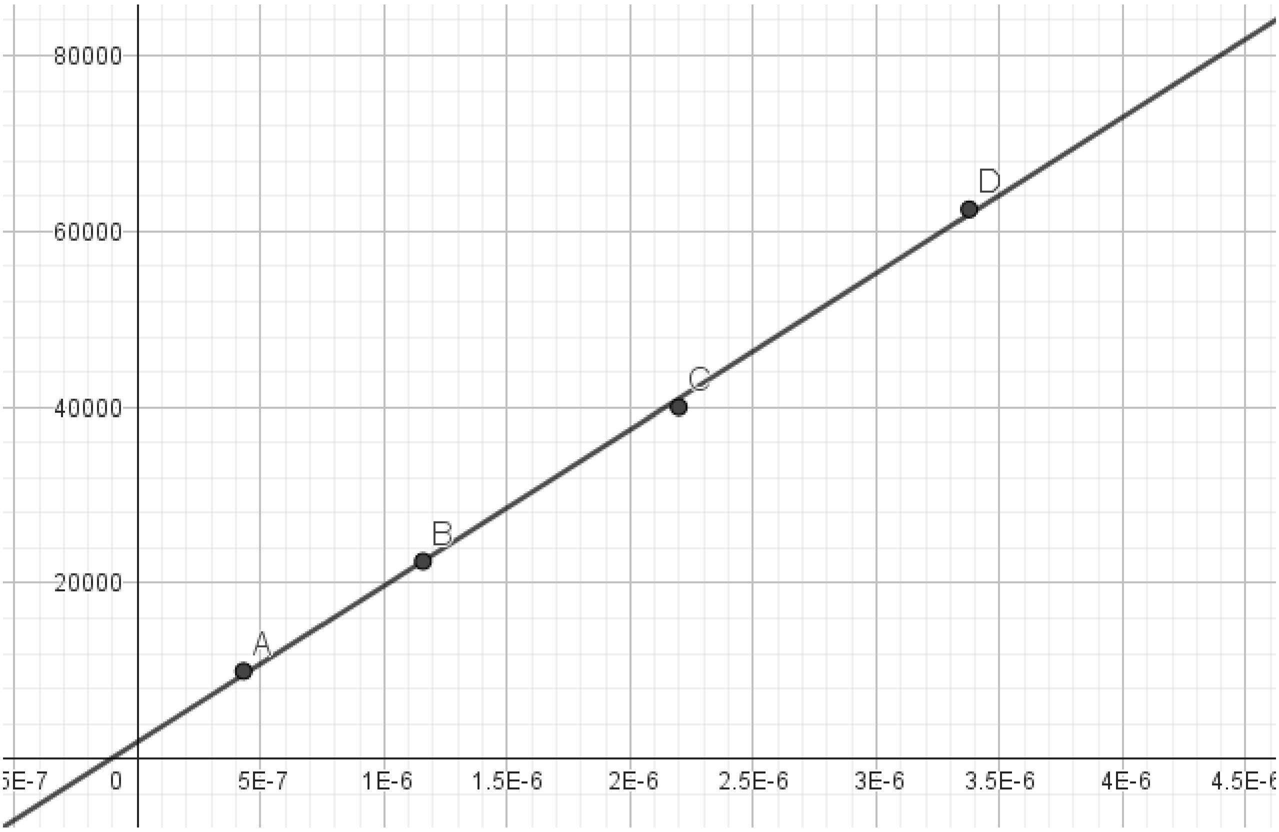
Using a tape measure, the following distance were measured
$D = 394\ cm = 3.94\ m$
$h_1 = 118\ cm$
$h_2 = 152\ cm$
$\therefore h = h_2 - h_1 = 36\ cm = 0.34\ m$

Wavelength of Laser Source:
$\Lambda = 635\ nm = 635 \times 10^{-9}\ m$
$\rho = 1000\ kg/m^3$

Now, for different values of frequencies set in the device, the following values of d were measured.

Frequency $f(Hz)$	$2d\ (cm)$	$d\ (m)$
100	1.55	0.00775
150	2.1	0.0105
200	2.6	0.013
250	3	0.015

Now plotting f^2 vs d^3, we get the following graph.



The Best fit line is $y = 1.8 \times 10^{10}x + 195$

$\therefore \sigma = 0.068\ N/m$